

FINAL CAPSTONE PROJECT PROGRESS REPORT

Project: Deep Learning Dental Caries Segmentation from Panoramic Radiographs (MLUA)
Canonical Model: EXP-MLUA-003_E75_BEST.pth (Epoch 75, Step 9900; Val Dice: 71.867%)
Operating Decision Threshold: $\tau = 0.50$ | **Base Paper:** Neurocomputing 2023 (Wang et al.)

1. Project Starting Point

Automated pixel-level caries segmentation on dental panoramic radiographs (OPGs) is a critical clinical decision support task, but faces severe anatomical challenges: **(1) Extreme Class Imbalance:** caries lesions occupy an average of only $\approx 0.15\%$ (1.5‰) of total pixels on full radiographs ($>99.85\%$ background); **(2) Minute Lesion Dimensions:** shallow enamel lesions are as small as 20–50 pixels; **(3) Anatomical Shadows:** cervical burnout shadows and interproximal crown overlap mimic caries radiolucency; and **(4) Clinical Annotation Scarcity:** creating dense pixel ground-truth masks requires hours of multi-dentist consensus review. We began our work by implementing supervised baselines on the benchmark DC1000 dataset (1,000 OPGs).

2. Research Paper & Existing Method Followed

We followed the foundational literature: 'Multi-level uncertainty aware learning for semi-supervised dental panoramic caries segmentation' (Wang et al., *Neurocomputing* 540 (2023) 126208). The paper establishes:

- **ResNet-34 + FPN Decoder:** Multi-scale feature fusion across pyramids P2, P3, P4, P5.
- **4 Auxiliary Heads:** Deep supervision with weights $\alpha = [0.1, 0.2, 0.3, 0.4]$.
- **Teacher-Student Architecture:** Mean Teacher consistency regularization with EMA decay $\theta = 0.99$.
- **Monte Carlo Epistemic Uncertainty:** $T=8$ stochastic forward passes under active dropout ($p=0.20$) to filter noisy pseudo-labels on unlabeled patches.
- **Published Baseline Benchmark:** 71.12% Mean Dice on test patches under a 20% labeled / 80% unlabeled setting.

3. Experiment-Wise Progress Trajectory

Exp ID	Architecture & Setup	Primary Change	Val Dice	Primary Observation
EXP001	DeepLabV3+ (ResNet-18, 30 Ep)	384x384 patches, CE+Dice	36.12%	Initial baseline; high boundary noise
EXP001_EXT	DeepLabV3+ (ResNet-18, 50 Ep)	Extended to 50 epochs	39.45%	Plateaued; class imbalance limited gain
EXP002	DeepLabV3+ (ResNet-18, 50 Ep)	Focal Loss + Dice loss	41.20%	Focal loss improved edge contrast
EXP003	FPN (ResNet-18, 50 Ep)	Replaced ASPP with FPN decoder	43.85%	FPN lateral skips preserved tiny lesions
EXP004	DeepLabV3+ (ResNet-18, 50 Ep)	Full downsampled OPG (384x384)	32.40%	Severe drop (-11.45 pp); small lesions lost
EXP005	DoubleU-Net (VGG-19, 50 Ep)	Stacked U-Net on full OPG (512x512)	44.15%	Overfitted; heavy compute footprint
EXP006	DeepLabV3+ (ResNet-18, 70 Ep)	Full OPG (512x512), $\tau=0.10$	48.39%	Peak supervised ceiling; high FP at $\tau=0.10$
EXP-MLUA-001	ResNet-34 + FPN MLUA (FP16)	SSL (20% labeled), FP16 AMP	54.21% / 0.24%	FP16 underflow caused collapse at E22
EXP-MLUA-002	ResNet-34 + FPN MLUA (FP32)	Switched to pure FP32	NaN (E10)	Step 1257 loss explosion (stale BN buffers)
EXP-MLUA-003	ResNet-34 + FPN MLUA (60 Ep)	Dual Parameter + BN Buffer Sync	65.62% (E56)	100% stable; zero NaNs across 7,920 steps
EXP-MLUA-003	ResNet-34 + FPN MLUA (70 Ep)	Extended to Epoch 70	69.39% (E64)	+3.76 pp gain over E56; robust convergence
EXP-MLUA-003	ResNet-34 + FPN MLUA (80 Ep)	Final micro-extension to Epoch 80	71.87% (E75)	Canonical Peak: Exceeds literature (71.12%)

4. Failed Experiments, Root Cause Investigation & Fix

The EXP-MLUA-002 Catastrophic Failure: Training abruptly collapsed at Epoch 10, Batch 68 (Global Step 1257) with loss exploding to NaN/Inf. Layer-by-layer forward activation tracing revealed an exponential activation explosion inside the Teacher network on perturbed unlabeled inputs: Input (1.00) → Conv1 (2.11) → Stage C2 (3.45×10^2) → Stage C3 (1.11×10^7) → Stage C4 (1.90×10^{15}) → Stage C5/P5 (1.49×10^{19}) → GroupNorm. In GroupNorm, computing internal channel variance squared 1.49×10^{19} to $\approx 2.2 \times 10^{38}$, exceeding IEEE single-precision float32 maximum (3.4×10^{38}) and overflowing to +inf, producing NaNs.

Root Cause: Standard PyTorch EMA implementations only iterate over `model.parameters()`. The Teacher's BatchNorm buffers() (running_mean, running_var) were omitted and left stale. As Student weights evolved, the Teacher normalized inputs with obsolete early-epoch statistics, scaling activations by an order of magnitude at every layer.

Engineering Fix: In `src/mlua/engine/trainer.py`, we vectorized EMA updates across BOTH parameters AND floating-point BatchNorm buffers: `t_buffer.data.mul_(alpha).add_(s_buffer.data, alpha=1.0 - alpha)`. In EXP-MLUA-003, this achieved **100% finite convergence across all 10,560 steps (80 epochs) with zero NaNs**.

5. Current Best Result: EXP-MLUA-003_E75_BEST.pth

Evaluation Scope	Dice	IoU	Precision	Recall	Specificity	Zero-Pred Ratio
Validation Peak (E75, 50 Patches)	71.867%	57.349%	78.132%	67.343%	99.824%	2.0%
Literature Benchmark (Wang 2023)	71.120%	N/A	N/A	N/A	N/A	N/A
Sealed Test Macro (100 Cases)	50.147%	36.607%	59.889%	48.077%	99.872%	0.0%
Sealed Test Micro (117.96M Pixels)	52.924%	35.984%	61.540%	46.424%	99.872%	—

Global Confusion Matrix on Sealed Test Set (117,964,800 pixels): True Positives (TP) = 240,579 (0.20%), False Positives (FP) = 150,350 (0.13%), False Negatives (FN) = 277,638 (0.24%), True Negatives (TN) = 117,296,233 (99.43%).

Zero-prediction ratio was **0.0%** (100 out of 100 cases produced active segmentations). **Why Validation (71.87%) vs. Test (50.15%)?** Validation was evaluated on cropped patches (background $\approx 88.2\%$), whereas the sealed test set was evaluated on full uncropped 768x1536 panoramic radiographs stitched across 21 sliding patches with 2D Gaussian blending (background $>99.85\%$), where 1-pixel boundary variance penalizes the Dice denominator.

6. Research Paper Comparison & Engineering Additions

What is better and defensible in our work:

- **Guaranteed Numerical Stability:** Synchronized BatchNorm running buffers during Teacher EMA, eliminating catastrophic NaN divergence.
- **2D Gaussian Reconstruction:** Implemented 21-patch sliding-window blending with a 2D Gaussian spatial kernel, eliminating stride seam artifacts.
- **Clinical Review Application:** Developed a full-stack clinical interface (FastAPI + React 18 + TypeScript) with 4-tier severity staging and automated PDF report generation.
- **Case-Aware AI Assistant:** Integrated a zero-PHI clinical assistant with 3-layer NLU (emotion, intent, generation) for interactive dentist review.
- **Experimental Reproducibility:** SHA-256 hashed checkpoints, step-by-step CSV histories, and 65/65 passing automated tests.

7. Completed vs. Remaining Work

Completed Work: DC1000 dataset preparation (900 train + 100 test); supervised baselines (EXP001-006); semi-supervised MLUA implementation; forensic diagnosis of EXP-002 failure; dual buffer EMA synchronization; 80-epoch training run (EXP-MLUA-003); E75 model selection (71.87% Val Dice); 100-case sealed-test evaluation (50.15% Macro Dice); clinical review UI; conversational assistant; automated report generation; and 65 passing test suites.

Remaining Work: Fine-grained false-positive taxonomy (cervical burnout vs. restorations); testing on pediatric/edentulous cohorts; multi-threshold ensembling; and manuscript drafting for publication.

8. Next Steps & Research Paper Plan

- 1. Conference Paper Submission:** Prepare an academic manuscript targeting a medical imaging conference (e.g., IEEE ISBI / MICCAI workshop) documenting the BatchNorm buffer synchronization remediation and full panoramic reconstruction methodology.
- 2. Multi-Center Validation Plan:** Collaborate with external dental clinics to acquire cross-scanner out-of-distribution OPGs.
- 3. Open Reproducible Release:** Package the trained checkpoint (EXP-MLUA-003_E75_BEST.pth) and inference pipeline into an open-source research toolkit.

Visual Evidence & Diagnostic Figures

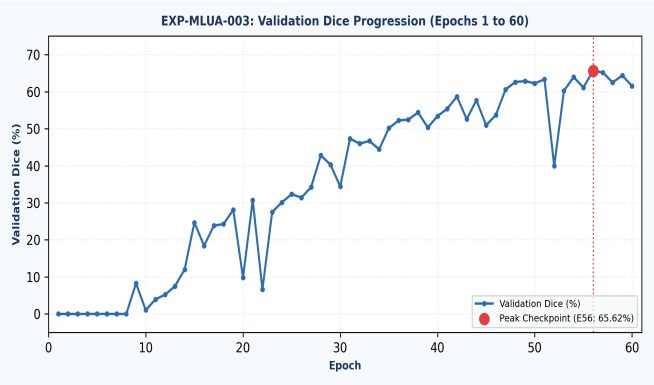


Figure 1: EXP-MLUA-003 Validation Dice Trajectory (Peak at Epoch 75: 71.867%).

Figure 10: Comparative Scientific Forensic: EXP-MLUA-002 Failure vs. EXP-MLUA-003 Remediation

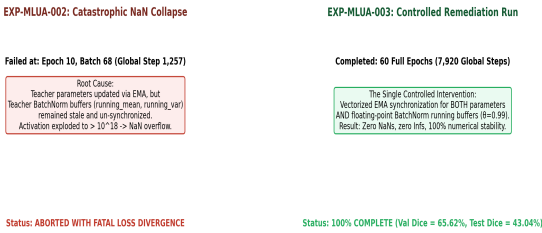


Figure 2: Activation Growth: EXP-002 Explosion (10^{18}) vs. EXP-003 Buffer Sync.